

MODEL QUESTION PAPER

REVISION-2010
SUB CODE:3088

Reg. No.....
Signature.....

THIRD SEMESTER DIPLOMA EXAMINATION IN POLYMER TECHNOLOGY
TECHNOLOGY OF ELASTOMERS

Time- 3 Hour

Max.marks-100

PART-A

(Answer *all* questions in one or two sentences. Each question carries 2 marks.)

- I
1. Define the term compounding.
 2. What is the role of peptiser in rubber compounding? Give two examples?
 3. Suggest suitable filler for 'green tyre'.
 4. What is 'scorch'? State method for avoid 'scorch'?
 5. How much phr of HAF required increasing 1 shore A hardness in NR and CR stock?

(2x5=10)

PART-B

(Answer any *five* questions. Each question carries 6 marks)

- II
1. a). What are the function of activators and retarders in a rubber compound?
Give examples. (3)
 - b). Write atypical recipe for a gum compound. (3)
 2. a). Explain CV, EV, and semi-EV vulcanization system. (3)
 - b). What are the non-sulphur vulcanization system. (3)
 3. a). Explain the importance of anti-degradents in rubber compound. (3)
 - b). Define plasticizers. Give its classification with examples. (3)
 4. a). Write down the procedure for determination of particle size of carbon black by chemical method. (3)
 - b). What are fibrous fillers, give examples. (3)
 5. a). Define cure time, scorch time, and optimum cure. (3)
 - b). Distinguish rate of cure and state of cure. (3)
 6. a). Define viscosity, elasticity, and plasticity of an elastomer. (3)
 - b). Name the various moulding techniques used in rubber product manufacturing. (3)
 7. What are the effect of particle size and structure of filler with respect to processing and vulcanisate properties. (6)

(5x6=30)

PART-C

(Answer *one* full question from each module; each question carries 15 marks)

MODULE -1

- III. a) List out the compounding ingredients used in a typical rubber compound.

- Explain its function with examples of each ingredient. (8)
- b) Explain the changes that occur in a rubber compound due to vulcanization. (7)

OR

- IV a) Explain the theory of sulphur vulcanization ? (8)
- b) Describe the chemical classification of accelerators with examples? (7)

MODULE –II

- V a) Explain the manufacture of furnace black with diagram? (8)
- b) List out the special purpose ingredients with examples? (7)

OR

- VI a) Explain the manufacture of thermal black with a neat diagram? (8)
- b) List out the properties of carbon black. Explain the chemical procedure for determining structure of carbon black ? (7)

MODULE –III

- VII a) Describe the various cure graphs of scorchy ,delayed action ,plateau cure Based on different type of accelerators (8)
- b) Explain the method of assessment rheological properties based on rheometer graph. (7)

OR

- VIII a) Explain different methods of batch vulcanization methods other than moulding. (8)
- b) Explain the rotocure vulcanization with neat diagram for the production of flat belt. (7)

MODULE –IV

- IX a) Describe the method of calculation of specific gravity and volume cost of a typical rubber compound. (10)
- b) What are the factors to be considered while designing tear resistant soft vulcanized rubber compound. (5)

OR

- X a) Design a NR based rubber compound d having a hardness of 60 shore A using silica as filler. Justify your choice. (8)
- b).What are the factors to be considered while designing abrasion resistant rubber compound. (7)
